Classification: Theory Questions and Answers

1. What is Logistic Regression and why is it suitable for classification?

Logistic Regression is a supervised machine learning algorithm used to predict the probability of a categorical outcome. It uses the sigmoid function to convert predictions into probabilities between 0 and 1.

2. Explain classification performance metrics and why accuracy alone is insufficient.

Classification metrics include Accuracy, Precision, Recall, F1-Score, and AUC-ROC. Accuracy alone can be misleading when the dataset is imbalanced because a model may predict the majority class correctly while failing to identify the minority class.

3. Define Type-I Error and Type-II Error in the context of risk prediction.

Type-I Error (False Positive): Predicting a person as high-risk when they are actually low-risk.

Type-II Error (False Negative): Predicting a person as low-risk when they are actually high-risk.

4. Explain Precision, Recall, F1-Score, TPR, and FPR.

Precision: Proportion of predicted positive cases that are actually positive.
Recall / TPR: Proportion of actual positive cases correctly identified.
F1-Score: Harmonic mean of Precision and Recall.
FPR: Proportion of actual negative cases incorrectly predicted as positive.

5. What is AUC-ROC and how does it help in evaluating classifiers?

ROC (Receiver Operating Characteristic) plots TPR against FPR at different classification thresholds. AUC (Area Under the Curve) measures how well a model distinguishes between positive and negative classes; a value closer to 1 indicates better discrimination.

6. Why does imbalanced data create problems in classification models?

Imbalanced data occurs when one class has significantly more samples than another. A model may become biased toward the majority class, resulting in poor detection of minority-class cases despite high accuracy.

Study Note: These questions and answers are based on the content provided.